CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

IO Organization

Lecture No.- 1



By- Vishvadeep Gothi sir

Recap of Previous Lecture



Topic Peripheral Device

Topic IO vs Memory Buses

Topic Asynchronous Data Transfer

Topics to be Covered



Topic IEEE-754 Floating Point Representation

Hexadecimal number system $(A16)_{16}$

microprocessor A16H

Programming 0xA16

[MCQ]

#Q. A certain well-known computer family represents the exponents of its floating-point numbers as "excess-64" integers; i.e., a typical exponent $e_6e_5e_4e_3e_2e_1e_0$ represents the number:

$(e_0 2^0 + e_1 2^1 + \cdots + e_6 2^6) - 64$

bias

A ✓ $e = -64 + \sum_{i=0}^{6} 2^i e_i$

B $e = -64 + \sum_{i=0}^{6} 2e_i$

C $e = 64 - \sum_{i=0}^{6} 2^i e_i$

D $e = 64 - \sum_{i=0}^{6} 2e_i$

[MCQ]

#Q. The value of a float type variable is represented using the single- precision 32-bit floating point format IEEE-754 standard that uses 1bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable X is assigned the decimal value of −14.25. The representation of X in hexadecimal notation is

A C1640000H

B 416C0000H

C 41640000H

D C16C0000H

[MSQ] [2025 : 2M]

#Q. Three floating point numbers X, Y, and Z are stored in three registers RX, RY, and RZ, respectively in IEEE 754 single precision format as given below in hexadecimal:

R_X = 0xC1100000, R_Y = 0x40C00000, and R_Z = 0x41400000

Which of the following option(s) is/are CORRECT?

A $4(X + Y) + Z = 0$

B $2Y - Z = 0$

C $4X + 3Z = 0$

D $X + Y + Z = 0$



Topic : Booth Multiplication

→ 2^s complement only

→ no. of operations are reduced.

234 *123 --- 702 468X 234XX --- 28782	1011 X101 --- 1011 0000X 1011XX --- 1101111



Topic : Booth Multiplication

Example:

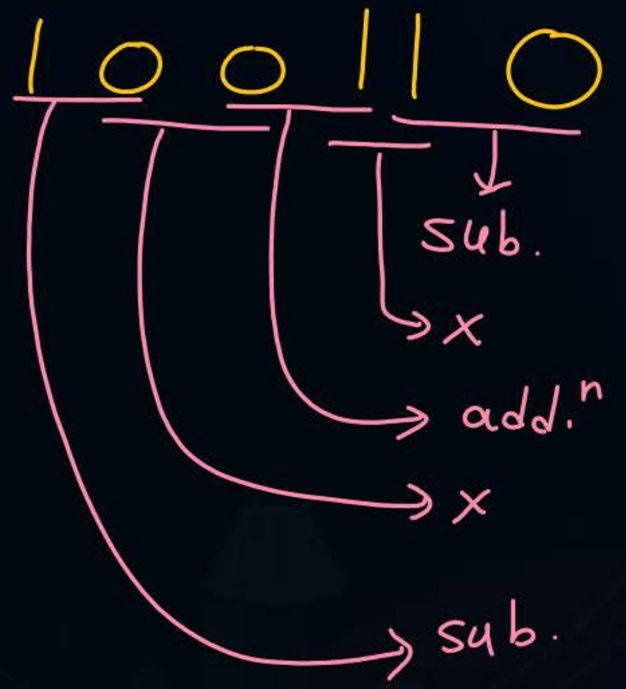
Multiplicand: 10111

Multiplier: 10011

1. Add a zero in right side of multiplier

10011 0

2. Check 2 consecutive bits { 10 ⇒ subtraction, 01 ⇒ Addition

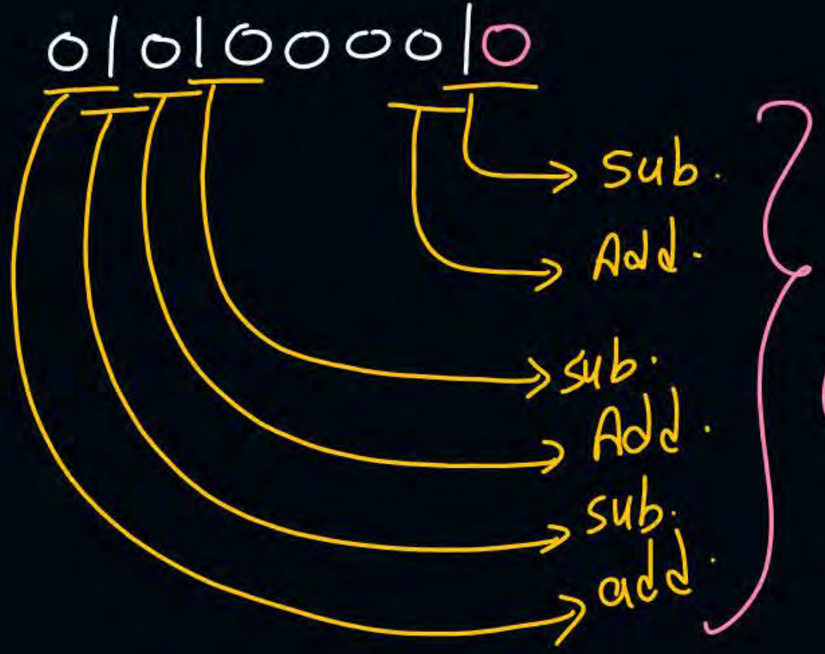


2 sub.
1 addition
} 3 operations

ex:-

1. multiplicand :- 11001001

multiplier :- 01010000



[NAT] [2025 : 1M]

#Q. The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm:

M: 1100 1101 1110 1101

and Q: 1010 0100 1010 1010 0

The total number of addition and subtraction operations to be performed is 13. (Answer in integer)

IO organization



Topic : Peripheral Device / I/O devices

All the devices which are connected to CPU externally, except main memory.

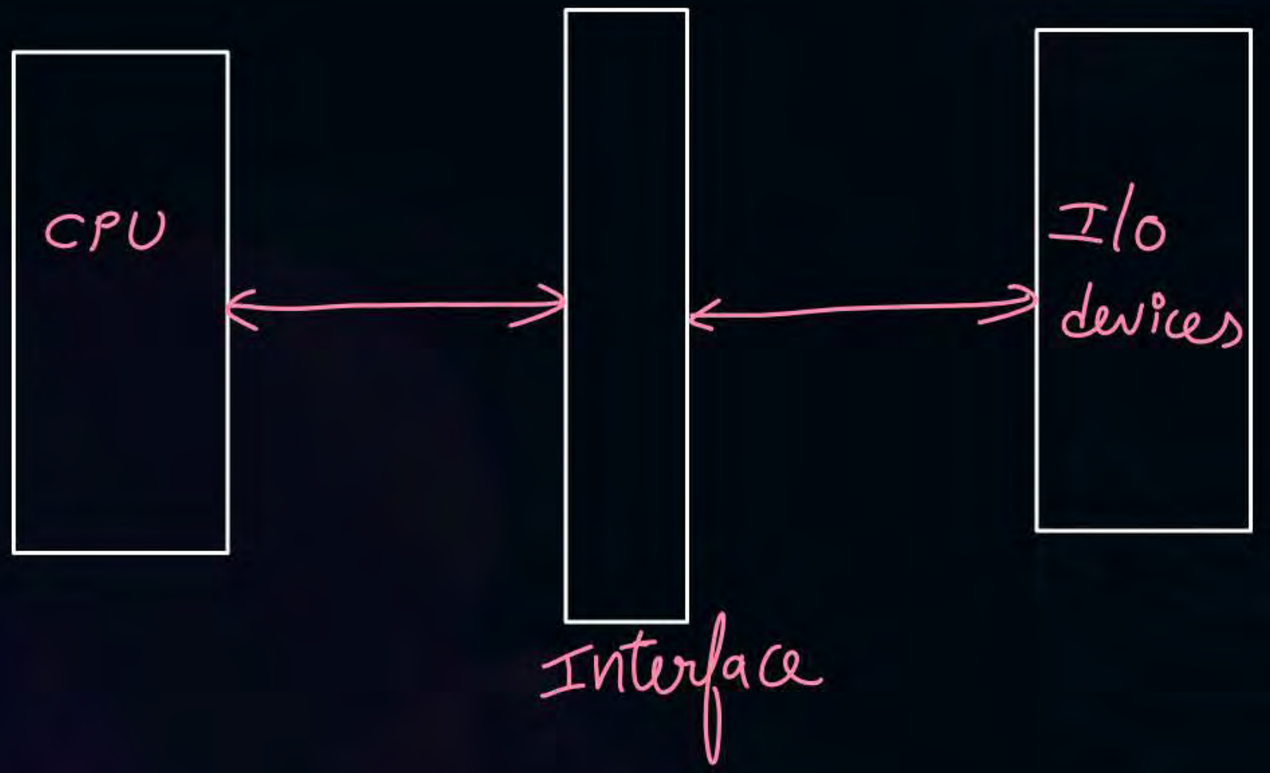
⟶ Input devices

⟶ output devices

⟶ storage devices



Topic : CPU Connected to IO Directly?





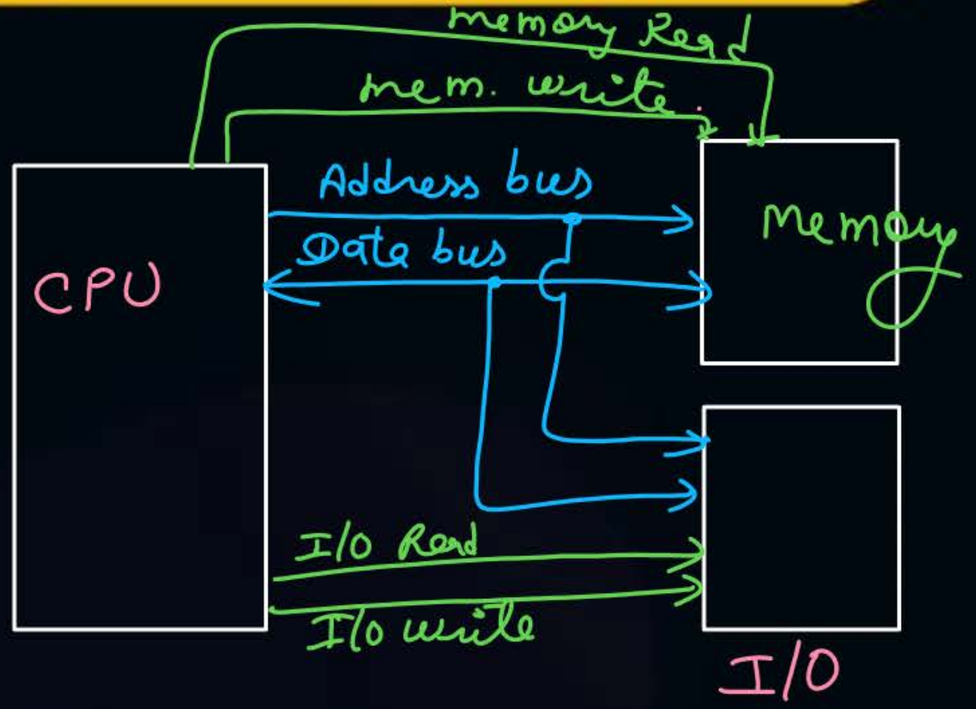
Topic : Need For Interface

1. Peripherals are electromechanical or electromagnetic devices; and their manner of operation is different from the operation of the CPU and memory. Which are electronic devices. So conversion of signal required.
2. The data transfer rate of peripherals is usually slow. So synchronization is required.
3. Data codes and format in peripherals differ from the word format in the CPU and memory. So conversion of formats is required.
4. The operating modes of peripherals are different from each other and each must be controlled so a peripheral does not disturb the operation of other peripherals.

I/O interface	Interfacing
DMA Controller	Interfacing + DMA
I/O Processor	—— 11 —— + DMA + I/O instns execution



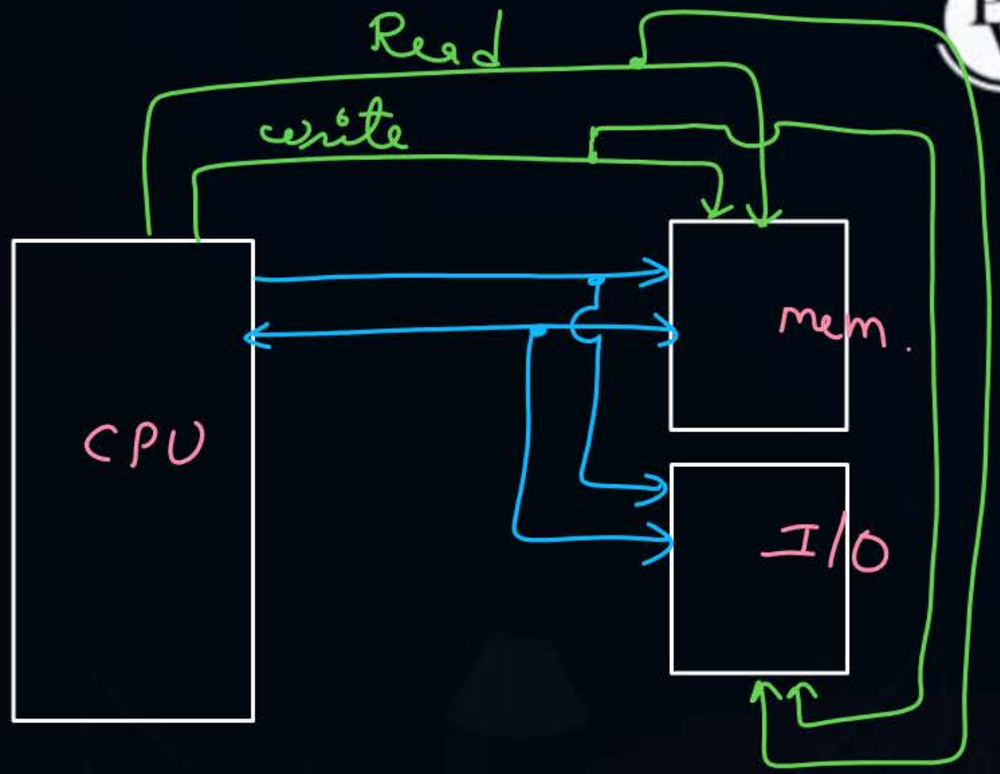
Topic : IO vs Memory Buses



I/O mapped I/O

(Port – II ——)

Isolated or I/O



memory mapped I/O



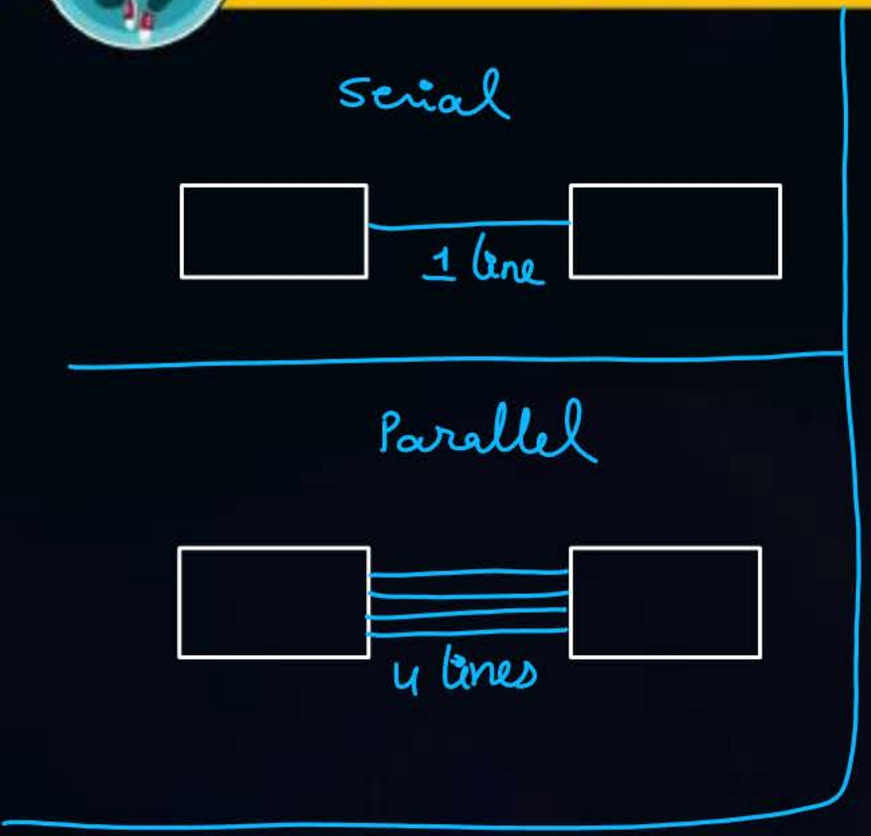
Topic : Memory Mapped IO vs IO Mapped IO

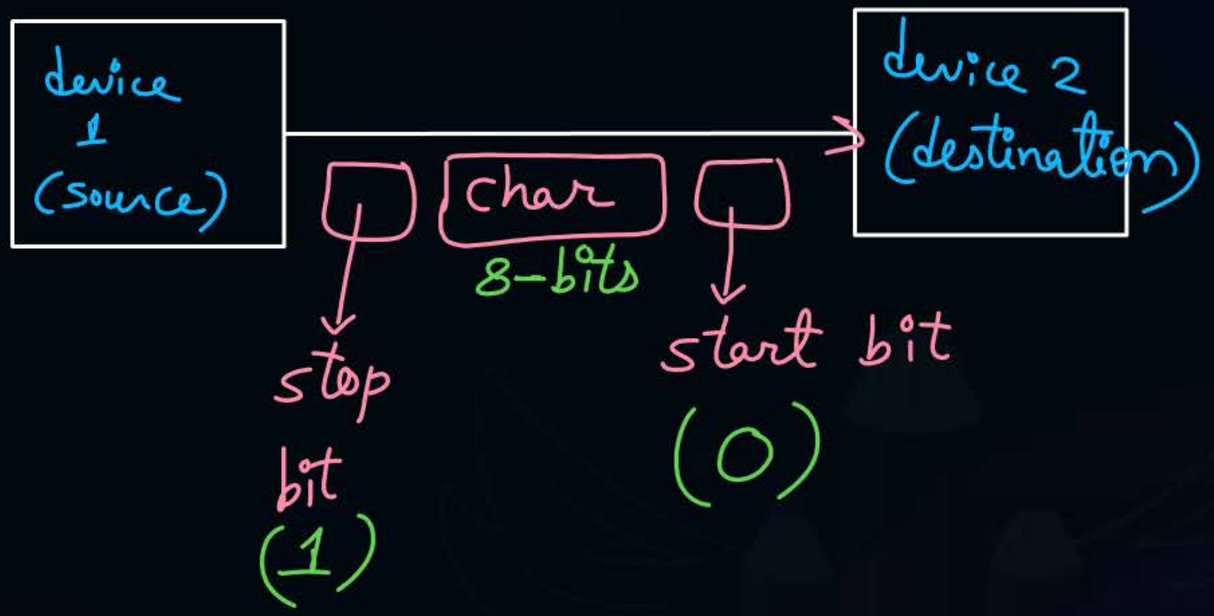
Memory Mapped IO	IO Mapped IO
1. Memory wastage	1. No Memory wastage
2. All Memory access instructions used for IO access also	2. IO access instructions and memory access instructions are different
3. No separate address space for IO	3. IO have their own separate address space
4. More Instructions for IO access	4. Less Instructions for IO access
5. More addressing modes for IO access	5. Less addressing modes for IO access
6. More devices can be connected to system	6. Less devices can be connected to system
7. ALU Operation can be performed on IO data directly	7. ALU Operation can not be performed on IO data directly



Topic : Asynchronous Data Transfer

serial





1. If no data transfer then destination sees '1' (stop bit)
2. As soon as destination observes first '0', that means data is coming.

To send each 8-bit char $\Rightarrow$ $8 + 1 + 1 = 10$ bits are transmitted.

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efficiency of line $= \frac{8}{10} = 0.8\ \%$

effective transmission rate = efficiency * Actual transfer rate

[NAT]

Ans = 800

#Q. How many 8-bit characters can be transmitted per second over 9600 ~~baud~~ bits/sec. serial communication link using a parity synchronous mode of transmission with 1 start bit, 8 data bits, 2 stop bits and 1 parity bit?

no. of bits per char = 1 + 8 + 2 + 1 = 12 bits

$$\text{char per sec.} = \frac{9600}{12}$$

$$= 800$$

[NAT]

#Q. An asynchronous serial communication is employing 8 character bits, 1 parity bit, 2 start bits and 1 stop bit. To maintain a rate of 650 char/sec the minimum transfer rate should be required is 7800 bits/sec?

1 char ⇒ 8 + 1 + 2 + 1 = 12 bits

650 char ⇒ 650 * 12 = 7800 bits

[NAT]

#Q. 8-bit characters can be transmitted using a parity synchronous mode of transmission with 1 start bit, 8 data bits, 2 stop bits and 1 parity bit.

1. What is the efficiency of the transmission line? $\frac{8}{1+8+2+1} = \frac{8}{12} = 0.67 = 66.67\%$
2. If the transfer rate of the line is 12000 bits per second, then effective transfer rate is?

$= \frac{8}{12} * 12000$

$= 8000$ bits/sec



Topic : Modes of Transfer

1. Programmed I/O or Program Controlled I/O
2. Interrupt I/O or Interrupt driven I/O
3. Direct Memory Access



Topic : Programmed IO

- There is no any provision through which IO can inform to CPU about data transfer
- IO sets its own status and waits
- CPU runs program periodically and checks the status of each device one-by-one
- If any device has its status set then CPU performs data transfer for it.

polling



total time required for data transfer in programmed I/O = status checking time + data transfer time

status checking time ⇓ depends on I/O speed



Topic : Interrupt Initiated IO

- IO device has a provision (Interrupt Signal) to inform to CPU about communication.



Topic : Interrupt Initiated IO

- IO device has a provision (Interrupt Signal) to inform to CPU about communication.
- When CPU receives interrupt:
  - It completes execution of current instruction
  - Saves the status (PC, PSW etc.) of current process onto the stack
  - Branches to service the interrupt
  - Resumes the previous process by taking out the values from stack

[MCQ] GATE – 2018

#Q. The following are some events that occur after a device controller issues an interrupt while process L is under execution.

P. The processor pushes the process status of L onto the control stack

Q. The processor finishes the execution of the current instruction

R. The processor executes the interrupt service routine

S. The processor pops the process status of L from the control stack

T. The processor loads the new PC value based on the interrupt

Which of the following is the correct order in which the events above occur?

A ✓ QPTRS

B PTRSQ

C TRPQS

D QTPRS



2 mins Summary

Topic Peripheral Device

Topic IO vs Memory Buses

Topic Asynchronous Data Transfer



Happy Learning

THANK - YOU